SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION with OpenCV

COURSE CODE: DSA0204

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CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

1. Object Boundary Detection

a. **Interpretation:** Accurately locating the boundary between different tissue types is essential for diagnosis, measurement, and treatment/surgical planning; the boundary corresponds to an intensity transition between adjacent tissue classes.

b. Key issues:

i. Low contrast between certain tissue types
ii. Noise in acquisition (e.g. MRI/CT artefacts)
iii. Blurred or gradual transitions rather than sharp edges
iv. Overlapping intensity ranges of different tissues

c. **Applied technique — Canny Edge Detection:** Apply Gaussian smoothing, compute intensity gradients, perform non-maximum suppression, then use hysteresis (dual) thresholding to trace connected edges.

d. **Justification:** Canny is well suited here because its built-in smoothing step reduces noise sensitivity, non-maximum suppression produces thin, well-localized edges, and hysteresis thresholding links weak-but-genuine edges to strong ones — giving accurate, continuous boundaries critical for medical use.

e. **Summary:** Requirement → accurate tissue boundaries; Fix → Canny edge detection; Result → precise, noise-robust boundary map for diagnosis.

2. Simple Segmentation

a. **Interpretation:** Since object and background intensities are well separated, this is a straightforward segmentation case with a clearly bimodal intensity distribution.

b. Key issues:

i. Histogram shows two well-separated peaks (bimodal)
ii. No significant overlap between object and background intensity ranges

c. **Applied technique — Global Thresholding using Otsu's Method:** Automatically compute the threshold that minimizes intra-class variance (equivalently maximizes inter-class variance) between the two intensity groups, then binarize the image at that value.

d. **Justification:** Because the histogram is clearly bimodal, a single global threshold is sufficient, and Otsu's method removes the need for manual/subjective threshold selection by choosing it statistically from the histogram itself.

e. **Summary:** Case → simple, well-separated intensities; Fix → Otsu global thresholding; Result → clean binary segmentation of object vs background.

3. Complex Segmentation

a. **Interpretation:** When several regions' intensity ranges overlap, no single threshold value can correctly separate all of them, making simple thresholding unreliable.

b. Key issues:

i. Overlapping histograms across multiple regions

ii. No single boundary value cleanly separates all classes **Clustering / Watershed Segmentation / Graph-based methods (e.g. Grab Cut):** Group pixels using multiple features (intensity plus spatial/gradient information) rather than a single global cut-off, or treat the gradient image as a topographic surface (watershed) to separate regions at ridge lines.

c. **Justification:** Unlike a single threshold, clustering and watershed-based methods use additional information (spatial proximity, local gradients, or feature vectors) to disambiguate pixels whose raw intensities overlap, correctly handling the complexity that simple thresholding cannot.

d. **Summary:** Case → overlapping intensities across regions; Fix → clustering/watershed segmentation; Result
→ correctly separated regions despite intensity overlap.

4. Broken Edges

a. **Interpretation:** Edge maps often contain gaps where the gradient magnitude briefly falls below the detection threshold (due to noise or a locally weak intensity change), even though the underlying boundary is continuous.

b. Key issues:

i. Strict/high threshold cutting off weak-but-genuine edge segments

ii. Noise causing local gradient dips along a true boundary

iii. Non-maximum suppression occasionally removing borderline pixels

c. **Applied technique — Edge Linking via Hysteresis Thresholding (Canny) or Morphological Closing:** Use two thresholds so weak edge pixels connected to strong ones are retained (hysteresis), and/or apply morphological dilation followed by erosion (closing) to bridge small gaps.

d. **Justification:** Hysteresis thresholding recovers genuinely-connected weak edges that a single strict threshold would discard, and morphological closing directly bridges small remaining gaps — together restoring continuity without introducing excessive noise-based edges.

e. **Summary:** Cause → gaps from thresholding/noise; Fix → hysteresis + morphological closing; Result → continuous, closed object boundaries.

5. Combined Requirement

a. **Interpretation:** The system has two competing goals at once: suppress sensor/lighting noise, while keeping the fine edges that indicate defects fully intact — a plain smoothing-then-sharpening approach risks reintroducing noise or losing detail.

b. Key issues:

i. Standard smoothing blurs the very edges needed for defect detection

ii. Standard sharpening amplifies any remaining noise

iii. Order of operations affects final quality

c. **Applied technique — Edge-preserving denoising (Bilateral filter or Median filter) followed by controlled sharpening (Unsharp Masking):** First denoise using a filter that respects edges, then apply a mild sharpening step to reinforce defect boundaries.

d. **Justification:** Bilateral/median filtering removes noise while explicitly preserving strong intensity discontinuities (edges); performing denoising before sharpening also ensures the later sharpening step does not amplify noise that a naive sharpen-first approach would.

e. **Summary:** Requirement → noise removal with edge preservation; Fix → denoise-then-sharpen sequence; Result → clean image with defect edges intact for reliable detectio

Code of Conduct:
I N. Harshavardhan (192324189) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Harshavardhan